

Supplemental information

**Clinically oriented prediction of patient
response to targeted and immunotherapies
from the tumor transcriptome**

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Supplementary Material

Supplementary Figures and Tables

Tuning cohorts						
Name	Source	Indication	Analyzable Drug	Background Therapy	N	N responders
<i>Bevacizumab</i>	GSE19860 ¹	Colorectal	Bevacizumab	FOLFOX	12	5
<i>Sorafenib</i>	GSE109211 ²	HCC	Sorafenib	None	67	21
<i>Lapatinib</i>	GSE66399 ³	Breast	Lapatinib	Chemotherapy	65	21
<i>Trastuzumab</i>	GSE50948 ⁴	Breast	Trastuzumab	AT followed by CMF	63	31
<i>BRAFⁱ</i>	GSE65185 ⁵	Melanoma	Vemurafenib/Dabrafenib	None	17	14
	GSE99898 ⁶				16	10
	GSE50509 ⁷				20	14
<i>Anti-PD1</i>	GSE91061 ⁸	Melanoma	Nivolumab	None	50	10
Evaluation cohorts						
Name	Source	Indication	Analyzable Drug	Background Therapy	N	N responders
<i>Bevacizumab₂</i>	GSE53127 ⁹	Colorectal	Bevacizumab	None	18	3
<i>Bevacizumab₃</i>	GSE103668 ¹⁰	Breast	Bevacizumab	Platinum	21	7
<i>Bevacizumab₄</i>	GSE60331 ¹¹	Colorectal	Bevacizumab	Chemoradiation	17	8
<i>Sorafenib₂</i>	GSE33072 ¹²	Breast	Sorafenib	None	39	20
<i>Trastuzumab₂</i>	GSE66399 ³	Breast	Trastuzumab	Chemotherapy	23	6
<i>Trastuzumab₃</i>	GSE37946 ¹³	Breast	Trastuzumab	Chemotherapy	50	27
<i>Trastuzumab₄</i>	GSE42822 ¹⁴	Breast	Trastuzumab	FEX/TX	25	12
<i>Trastuzumab₅</i>	Sammut et al. ¹⁵	Breast	Trastuzumab	FEC	65	19
<i>Cetuximab</i>	GSE65021 ¹⁶	H&N	Cetuximab	Platinum	40	14
<i>Selinexor</i>	GSE186332 ¹⁷	GBM	Selinexor	None	24	8
<i>MK2206</i>	GSE150576 ¹⁸	Breast	MK2206	None	20	6
<i>Tipifarnib₁</i>	GSE5122 ¹⁹	AML	Tipifarnib	None	57	13

<i>Tipifarnib₂</i>	GSE8970 ²⁰	AML	Tipifarnib	None	34	13
<i>Rituximab</i>	GSE35935 ²¹	CLL	Rituximab	Chlorambucil	62	16
<i>Alpelisib/Ribociclib</i>	This manuscript	Breast	Alpelisib/Ribociclib	None	28	16
<i>Anti-PD1₂</i>	Zhao et al. ²²	GBM	Nivolumab/Pembrolizumab	None	15	9
<i>Anti-PD1 +- Anti-CTLA4</i>	GSE140901 ²³	HCC	Nivolumab/Nivolumab + Ipilimumab/PDR001 + MBG45	None	9	5
<i>Anti-PD1₃</i>	GSE67501 ²⁴	RCC	Nivolumab	None	11	4
<i>Anti-PD1₄</i>	GSE173839 ²⁵	Breast	Durvalumab	Olaparib	71	29
<i>Anti-PD1₅</i>	Cui et al. ²⁶	Melanoma	Anti-PD1 (drug undisclosed)	None	55	14
<i>Vismodegib</i>	GSE58375 ²⁷	BCC	Vismodegib	None	13	4

Table S1. Cohorts used for tuning (top) and evaluation (bottom), Related to Figures 1,2. Name: Name as appears in main text. Source: The source from which the datasets were obtained. All datasets are also available in <https://github.com/PangeaResearch/enlight-data>. Analyzable Drug: the drug used for ENLIGHT score generation. ‘A/B’ indicates a patient received either A or B. ‘A +- B’ indicates that some patients received A and some received both A and B. Background Therapy: treatments used in combination with the analyzable drug that were not considered when calculating the EMS. N: number of patients analyzed. N responders: number of patients classified as responders.

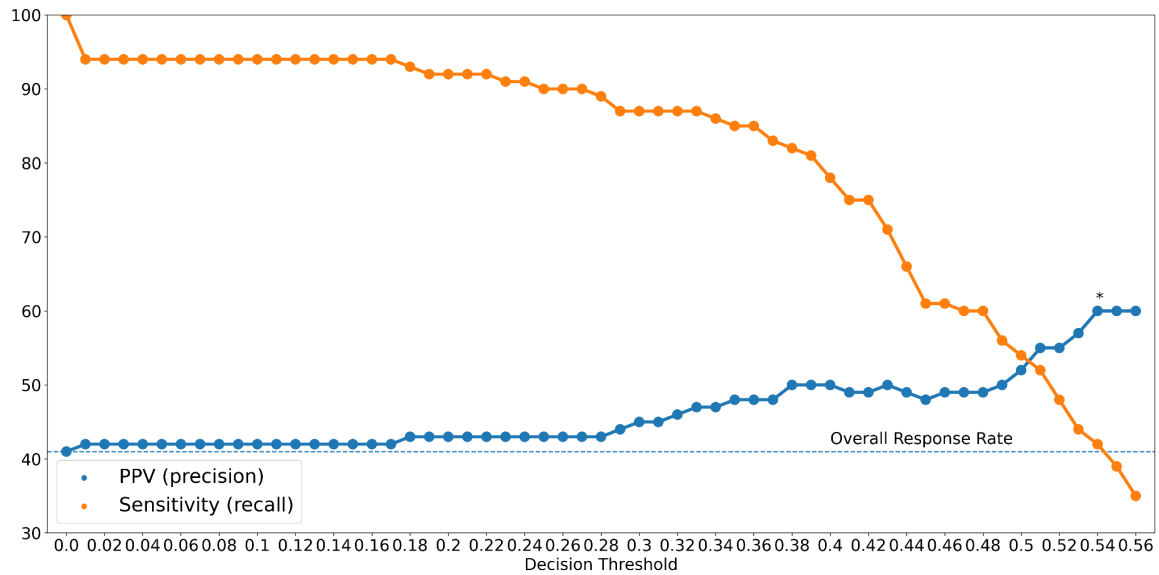


Figure S1. PPV (precision) and sensitivity (recall) for EMS on the tuning cohorts, Related to Figure 1. The y axis displays the PPV or sensitivity as a function of the decision threshold shown on the x axis. Figure depicts decision thresholds for which there was at least 30% recall.

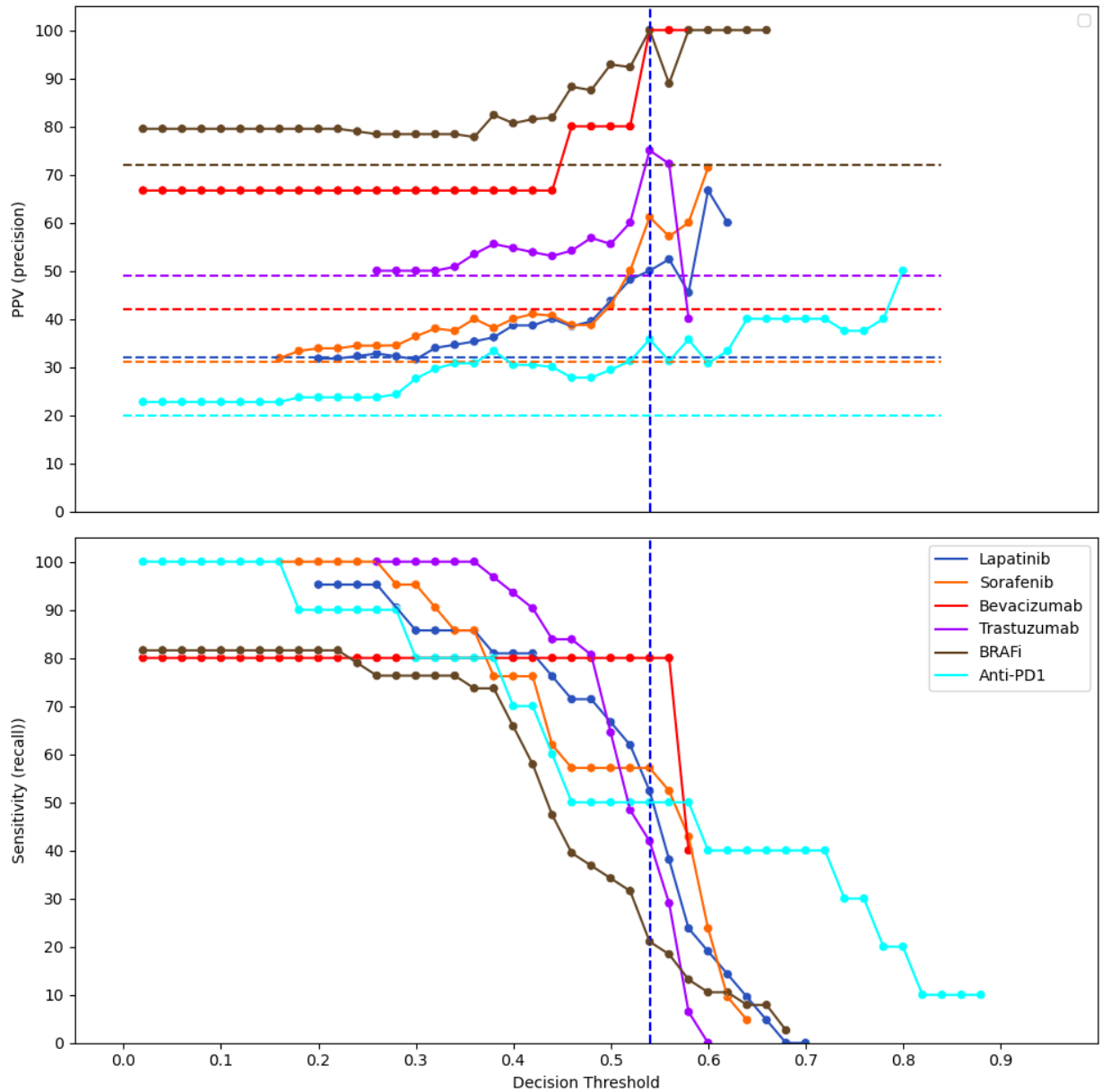


Figure S2. PPV and Recall as a function of the decision threshold, Related to Figure 1. The PPV (precision) (top) and sensitivity (recall) (bottom) are presented separately for each of the tuning cohorts. Horizontal lines of corresponding color depict the baseline PPV, i.e., the overall response rate in the cohort.

	N	Response Rate	PPV (precision)	OR
Targeted Small Molecules	215	37%	45%, $p=0.018$	1.59 [0.74, 2.45], $p=0.094$
ICB	161	38%	49%, $p=0.0008$	2.39 [0.98, 4.84], $p=0.007$
mAb	321	39%	57%, $p=2.37e-10$	3.72 [2.31, 5.98], $p=2.24e-7$
All	697	38%	52%, $p=3.30e-13$	2.59 [1.85, 3.6], $p=3.41e-8$

Table S2. ENLIGHT Performance by Therapeutic Class, Related to Figure 2. PPV (precision):

The percent of patients identified as matching a treatment that indeed responded. The p-values for the difference between PPV and response rate were calculated using the one sample proportion test. Odds Ratio: The odds ratio for response of ENLIGHT-matched cases; Square brackets indicate the 95% confidence interval; The p-value is calculated using Fisher's exact test.

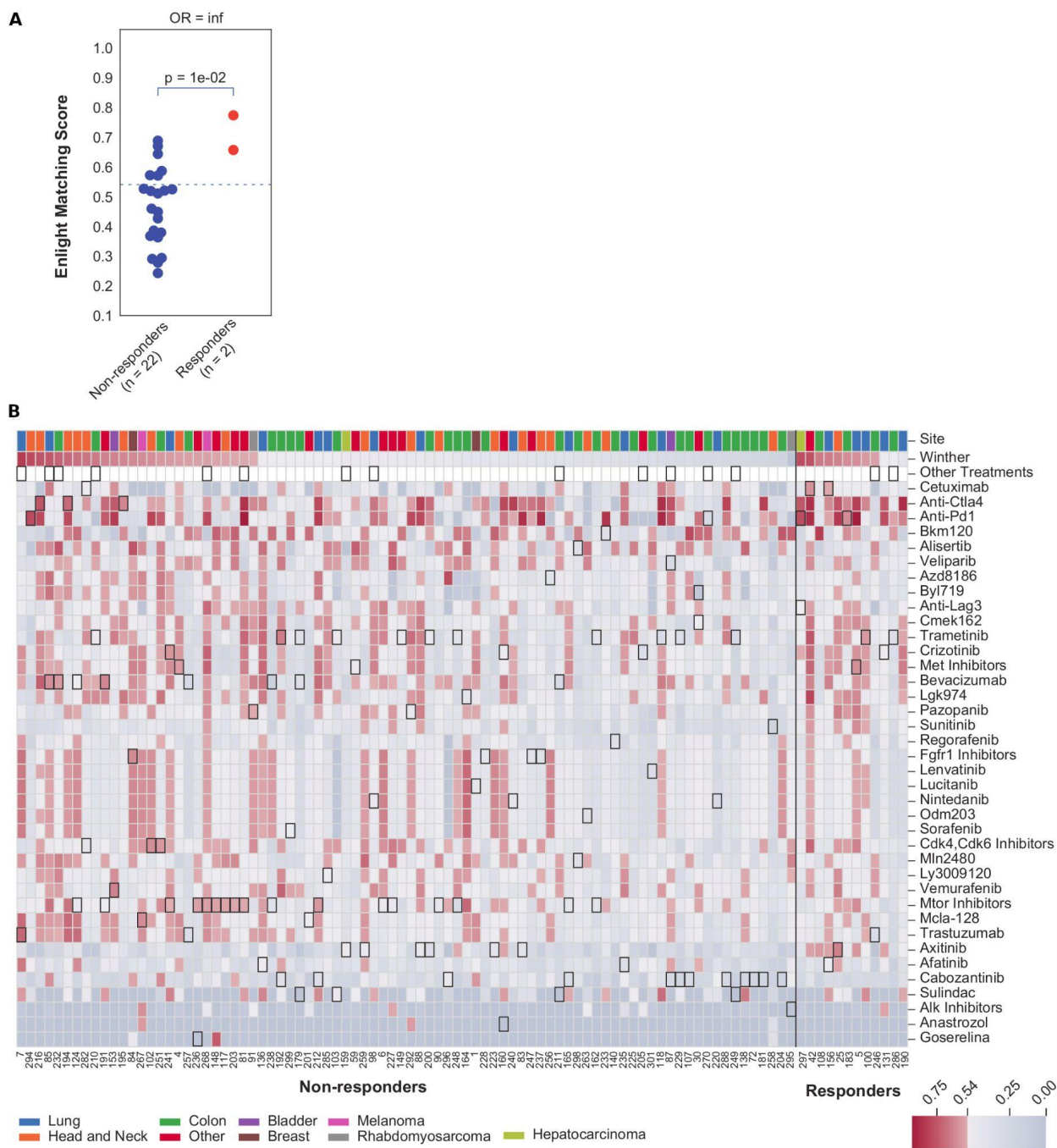


Figure S3. Further analysis of the WINTHER trial, Related to Figure 2. (a) Analysis of the 24 patients that were treated with a combination of ENLIGHT-analyzable drugs in the WINTHER trial. Responders (orange) have significantly higher EMS than non-responders (blue), p-value is based on one sided Mann-Whitney test. The horizontal line marks the decision threshold for considering a treatment as favorable for a patient ($EMS \geq 0.54$). OR: odds ratio

for response for patients receiving treatments with an EMS above the decision threshold.

(b) The heatmap shows the EMS for the 96 patients analyzed in the WINTHER trial (columns) and all ENLIGHT analyzable drugs given in the trial (rows). The 'Winther' row shows the EMS for the treatment regimen given in the trial. Color designates EMS, with red colors corresponding to ENLIGHT-matched treatments ($EMS \geq 0.54$). Black boxes indicate the drugs that were given to each patient. 'Other treatments': non-analyzable drugs, i.e., chemotherapy or hormonal therapy. The cancer type of each sample is color-coded at the top of the heatmap.

	N	OR
All	96	11.15 [2.28, 54.54], $p=7.80e-04$
ENLIGHT Analyzable Only	81	10.20 [1.99, 52.24], $p=2.48e-03$
ENLIGHT Analyzable + Unanalyzable	15	inf
Monotherapy ENLIGHT Analyzable Only	60	8.14 [1.47, 45.18], $p=0.01$
Monotherapy ENLIGHT Analyzable + Unanalyzable	12	inf
Combination ENLIGHT Analyzable Only	21	inf
Combination ENLIGHT Analyzable + Unanalyzable	3	NA
Monotherapy	72	8.40 [1.63, 43.24], $p=6.07e-03$
Combination	24	inf

Table S3. ENLIGHT predictions on combination therapies in the WINTHER trial, Related to Figure 2. Patients in the WINTHER trial received either a single drug (monotherapies) or a combination of drugs. Moreover, some of the patients received drugs unanalyzable by ENLIGHT, i.e., chemotherapies or hormonal therapies. N denotes the number of patients; in

square brackets - 95% confidence interval for the odds ratio; the p-value is calculated using Fisher's exact test.

SELECT sets, N = 297 (Figure 3b)				
Biomarker	OR	p	95% CI	p value under H0: ENLIGHT does not have a greater OR than the respective marker (one sided)
ENLIGHT	2.308	0.0004	[1.42,3.73]	
SELECT	1.168	0.154	[0.72,1.875]	0.0487
non-ICB targeted therapies, N = 512 (Figure 3c)				
Biomarker	OR	p	95% CI	p value under H0: ENLIGHT does not have a greater OR than the respective marker (one sided)
ENLIGHT	2.71	1.32E-06	[1.24,4.59]	
gm(ENLIGHT,IFNG)	2.163	0.002	[1.4,3.341]	0.233
Proliferation	0.911	1	[0.636,1.304]	1.22E-05
IFNG	1.392	0.064	[0.962,2.014]	0.006
Cytolytic	1.44	0.064	[1.003,2.002]	0.007
gm(IFNG,Cytolytic,Proliferation)	1.33	0.073	[0.927,1.912]	0.003
Target Expression	1.399	0.064	[0.977,2.002]	0.005
Random Genes	0.773	1	[0.531,1.125]	1.72E-06
ENLIGHT-InVitro	0.824	1	[0.560,1.212]	9.14E-06
Other Drugs SL	0.949	1	[0.622,1.448]	0.0002
ICB datasets, N = 152 (Figure 3d)				
Biomarker	OR	p	95% CI	p value under H0: ENLIGHT does not have a greater OR than the respective marker (one sided)

ENLIGHT	2.69	0.037	[1.24,4.59]	
gm(ENLIGHT,IFNG)	4.076	0.002	[1.946,8.539]	1
SELECT	1.798	0.094	[0.904,3.57]	0.208
ENLIGHT-InVitro	0.832	1	[0.409,1.689]	0.01
Target Expression	2.111	0.088	[1.06,4.204]	0.312
Other Drugs SL	0.77	1	[0.365,1.722]	0.01
Random Genes	1.031	0.272	[0.524,2.027]	0.024
Proliferation	2.26	0.094	[0.98,5.213]	0.386
Cytolytic	2.342	0.092	[1.041,5.270]	0.406
gm(IFNG,Cytolytic,Proliferation)	2.039	0.088	[4.018,6.459]	0.285
IFNG	2.543	0.037	[1.291,5.012]	0.454
Exhausted T-cell	1.733	0.119	[0.882,3.404]	0.183
TIDE	1.144	0.271	[0.586,2.234]	0.038
CD8+ T-cell	1.622	0.139	[0.831,3.167]	0.147

Table S4. OR of ENLIGHT and other biomarkers, Related to Figure 3. Each sub-table contains the OR, the p value for a test of the OR being greater than 1, the 95% CI of the OR and the p value for a test of greater OR for ENLIGHT vs. the corresponding biomarker (one sided).

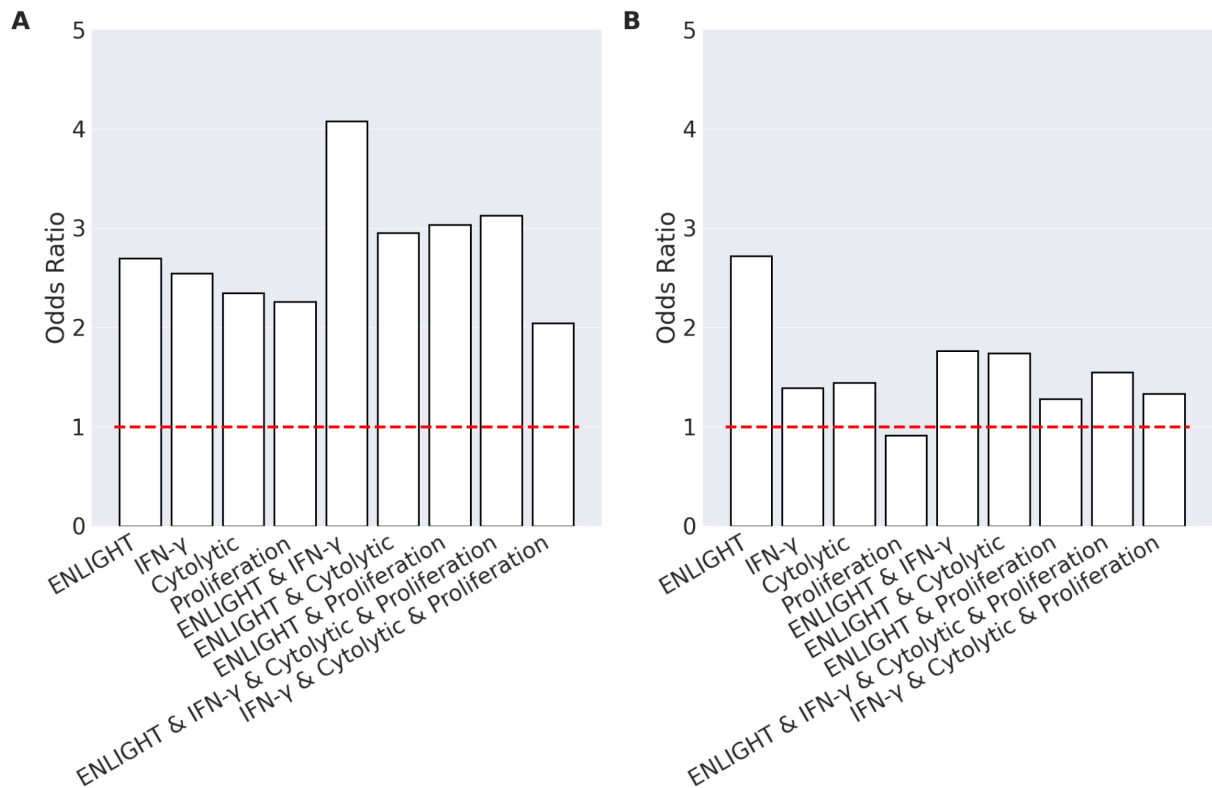


Figure S4. Comparison of OR between ENLIGHT and IFN- γ , Cytolytic and Proliferation signatures along with the combination between ENLIGHT and the three signatures, Related to Figure 3. Comparison is given for ICB datasets (**a**, N= 152) and non-ICB datasets (**b**, N = 511). X & Y refers to the geometric mean between the X and Y signatures per patient. For each signature, the decision threshold was calibrated as described in **STAR METHODS** (except *ENLIGHT* where the 0.54 threshold was used). The red dashed line represents an OR of 1 which is expected by chance.

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